DOMAIN GUIDE

Participant Access And Loan Operations

This domain turns the separate agreement, Vault, Base, and Bitcoin authorities into a usable product workflow. It records who may act, what the application durably accepted, which public operation is pending, and which externally sourced result is known. It also owns a deliberately isolated developer demonstration. Application records coordinate those authorities; they cannot become private keys, participant consent, or chain truth.

> This is the domain guide for Participant Access And Loan Operations, one of the five domains of Bitcoin-backed USDC one-loan system. The architecture guide, a separate document, explains how the domains work together.

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A **workspace** is the durable application view of a loan. A **revision** identifies the exact state a command was based on. An **outbox** records required work atomically with the state change so a lost immediate wake cannot lose the operation. A **handoff** grants a particular participant access to one bounded Vault action; it is an expiring pointer, not authentication, transaction consent or cryptographic authority. A **payment intent** fixes one exact public request before the participant's Vault signs. These concepts distinguish an accepted request from a submission, observation, finalized contract event, or confirmed Bitcoin spend.

Access, Coordination, And Local Composition

Participant identity and role access owns accounts, role membership, sessions, recovery, reauthentication and multi-factor authentication (MFA), and permission to retrieve bound opaque artifacts. Product authentication uses the selected provider adapter while stable accounts and loan authorization remain application-owned. Workforce operator access is separate. Authentication can authorize ciphertext retrieval; it cannot decrypt a Vault, recover a seed or secret, or make a chain action valid.

The identity, session, and fresh access states and the access contracts retain exact denial and recovery behavior.

Loan workspace and application coordination owns revisioned commands, idempotency, durable outbox work, source records, projections, notifications, package archives, sponsorship records, and exact participant handoffs. A consumer can acquire and refresh its authorized workspace, distinguish absent, forbidden, stale, and unavailable results, obtain an accepted-command acknowledgement, and recover the same result after refresh or restart. A mock, health route, or producer-local method does not prove that public consumer path.

The workspace command contract and the workspace states preserve the exact routes and failures.

Local runtime composition and controlled demonstration assembles the existing independently built units through their public seams. Its one developer entrypoint preflights, starts or resumes, health-gates, and reports the system; paired status, shutdown, and explicit reset/reseed actions make state ownership clear. The local profile uses a local Base test chain (Anvil), fake USDC, two connected persistent Bitcoin nodes in regression-test mode (regtest), two isolated participant sessions/Vaults, the real Product/Vault/API/observer/operations paths, and identity-bound fixtures. A coordination descriptor references producer-owned state instead of owning or recursively deleting sibling repositories. Ordinary shutdown preserves that state; only explicit bounded reset invokes each producer's reset contract.

The local demo runtime control contract and the local demo runtime states own the lifecycle.

Cooperation Through A Loan

The application service, called Core, first binds authenticated participants to a reserved loan and current terms candidate. The dedicated Vault returns acknowledged public identity and preparation material; under the terms domain's detailed rules both roles approve the same final candidate, which is one side of an unresolved source conflict over whether the lender approves again after reservation, explained in the loan terms guide. Bitcoin operations produce complete unsigned signing context, Core stores an immutable batch, Product launches the bounded Vault handoff, and the Vault signs locally. Core revalidates the result, writes the full package archive, and only then issues its receipt. After qualified funding and deliberate borrower acceptance, observed Base `X` and independent Bitcoin evidence drive separate projection and operation steps. Normal repayment similarly waits for finalized contract outcomes and confirmed Bitcoin recovery. The origination flow, the activation flow, and the normal repayment flow provide the authoritative sequence.

A required state change and its job commit together in the D1 workspace database. Immediate private wake is only an optimization; a one-minute scan recovers missed or crashed work. Exact request, route, participant, amount, nonce, fees, bytes, provider, and transaction identities survive ambiguous outcomes. Consumers do not turn a lost response into a new payment or changed action. Known inclusion delay remains distinguishable from unresolved reconciliation; the bounded policy probes at 0, 15, and 60 seconds and escalates at 180 seconds to the declared visible block/recovery route.

The Vault-owned payment account and exact transfer contract and the payment account and intent states retain the exact phase, fencing, and evidence contract.

Payment Authority And Availability

A Vault payment account is independent of its protocol externally owned account (EOA) and per-loan Bitcoin keys. It becomes public only after snapshot and backup recoverability establish same-address restore. Core owns the public intent, lease revision, fence, atomic phase transitions, and provider gate; the Vault owns encrypted pending bytes and signing. Core enters durable Signing before invoking the signer. Public hash and broadcast-phase persistence must occur in their specified order; repeat broadcast uses identical bytes, and changes require fresh review. A timeout after Signing is not proof of non-submission. A new device must restore or verifiably reproduce the exact pending bytes, otherwise the account remains explicitly blocked. None of this changes historical terms, payer-neutral settlement, or observation authority.

The selected narrow participant-transfer purpose permits configured USDC to a validated user-selected recipient with exact approval and balance/gas checks. It has no invented loan/workspace binding and cannot advance loan state. Arbitrary calls, approvals, swaps, other assets, native-value sends, and generic signing remain excluded. Local setup may provision disposable funds to public payment addresses, but that is neither participant-payment evidence nor production gas sponsorship. The latter remains an excluded ticketed capability.

All selected V1 application artifacts share Cloudflare's provider failure domain, despite separate origins, projects, credentials, and releases. A provider-wide outage can suspend the Product, Vault, API/D1, observers, and automation together. The chains continue; new consequential application commands stop. Return requires canonical Base/Bitcoin reconciliation before idempotent jobs resume. Off-provider archive backups protect data recovery, not a running-service fallback or promised recovery time or recovery point objective (RTO/RPO).

Recorded Choices And Scope

The public-consumer continuity work was needed because a producer-local command and passing harness alone do not ensure a browser can acquire state or recover an accepted operation. The local composition was added because independently implemented units did not by themselves supply a persistent runnable demonstration or safe profile separation. Vault-owned payment intent prevents unrecoverable exposed addresses, duplicate payment after lost responses, and accidental mixing of protocol keys with payer identities. The exact mechanics stay with their contracts; prior implementation repair history stays with tickets and handoffs.

Verification, runnable local use, testnet/mainnet deployment, and activated production operation remain distinct claims under the runtime profile claims and activation gates. Payment accounts remain a selected design with unconfirmed implementation and verification status (see Open questions).

Forgotten-password recovery and complete cooperative-settlement operation remain deferred under the stronger participant-Vault recovery and cooperative settlement operations exclusions. A visual, fixture, healthy process, or local design check cannot satisfy those exclusions or activation gates.

The existing future provider-replacement question remains deferred: WorkOS is the selected V1 identity provider, and a replacement must preserve internal account and immutable loan/Vault identities while proving authenticated linking, session/MFA/recovery parity, cutover, rollback and adapter conformance (application-owned authentication replacement).

Optional platform-paid reveal relaying/paymaster execution is also deferred; current support is bounded protocol-EOA prefunding plus participant signing and direct submission (platform-paid Base reveal execution). These are existing exclusions, not newly selected capabilities.

Open questions

Payment status is unsettled: the payment account selection leaves implementation and execution evidence gated, and browser acceptance verification calls the capability planned and unverified. Handoff records describe implementation and focused checks without finalized completion receipts. The domain record retains that reconciliation; this guide does not claim an available verified capability.

Authentication requirements need reconciliation. The identity-state contract makes MFA optional to enroll in, then mandatory for enrolled participants' sensitive actions. The repayment flow lists fresh reauthentication/MFA without that qualification. This guide does not choose universal or enrollment-dependent MFA.

The architecture guide shows where this domain fits in the system, and the domain record lists exactly what the domain owns and excludes.

Other guides: Loan Terms And Commitments.